Alyssa Brittany Chen

University of California, Berkeley

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GitHub
LinkedIn

Education

University of California, Berkeley *2022–2026*

B.A. Data Science, Applied Math Concentration

Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics & Probability Theory, Multivariable Calculus, Quantitative Finance, Optimization Models, Artificial Intelligence, Machine Learning.

Experience

Quantitative Research Associate | Fidelity Investments *June 2026*

– Researching and developing quantitative strategies to improve Fidelity Digital Asset crypto products.

Investment Intern | Blockchain Capital *Sept 2025 - Dec 2025*

– Developed a multi-sheet automated dashboard integrating multiple crypto data APIs to aggregate sector specific crypto metrics (e.g. quarterly growth rates, P/S ratios), enabling the investment team to quickly diligence performance across categories for portfolio companies and liquids.
– Synthesized market research on tokenized public and private equities to lead a 2-hour investment discussion, evaluating market structure, regulatory risk, and thesis viability for investment diligence.
– Sourced and led 20+ investment pitch calls.

Research Associate Intern | Fidelity Investments *Jun 2025 - Aug 2025*

– Developed and deployed ARIMAX time series models in Python (pandas, statsmodels, scikit-learn) to forecast staking yields for Solana and Ethereum, achieving an R^2 of 0.80 to improve projecting future revenue generation for Fidelity's ETP staking products.
– Engineered automated Python pipelines leveraging Snowflake SQL, CoinGecko API, and Token Terminal API to extract top tokens by date via index weights, leading to a more streamlined process of portfolio index maintenance and re-balancing.
– Conducted risk assessments on blockchain projects, producing insights to inform investment strategies and portfolio positioning.

Blockchain at Berkeley | Head of Education *Aug 2024 - May 2025*

– Increased organization visibility by writing and editing educational X/Twitter threads, generating 50k+ impressions across four research-focused threads.

Research

Threshold Cryptography in Enhancing the Security and Scalability of DeFi *2023*

– Investigates the integration of threshold cryptographic techniques in DeFi platforms to address security and scalability using multiparty computation, Boneh Lynn Shacham (BLS) scheme, and FROST (Flexible Round-Optimized Schnorr Threshold Signature Scheme) protocol.

Personal Projects

Statistical Arbitrage Model *Pandas, NumPy, Statsmodels, Scikit-learn, YFinance, Matplotlib*

– Engineered a quantitative pairs trading strategy using z-score-based mean reversion signals, cointegration testing, and volatility scaling to capture relative-value opportunities across equity pairs.
– Applied time series analysis and statistical arbitrage techniques to model long/short positions with dynamic risk controls, achieving a 55% win rate and Sharpe ratio optimization through signal threshold tuning.

The Color Theory *React, Supabase, Tailwind, Framer, Hume API, OpenAI, ngrok*

– Built a system that takes in audio transcripts from Omi and inputs it into a general-purpose LLM that's integrated with an MCP server via the Hume API. It translates them into color personality profiles based on emotional tone, sentiment, and behavioral traits.

Technical Skills and Interests

Languages & Frameworks: Python, Java, Javascript, SQL, Rust, Solidity, React, NumPy, Pandas, PyTorch, R, MATLAB.